UNIT 2 · TOPIC 10(A)

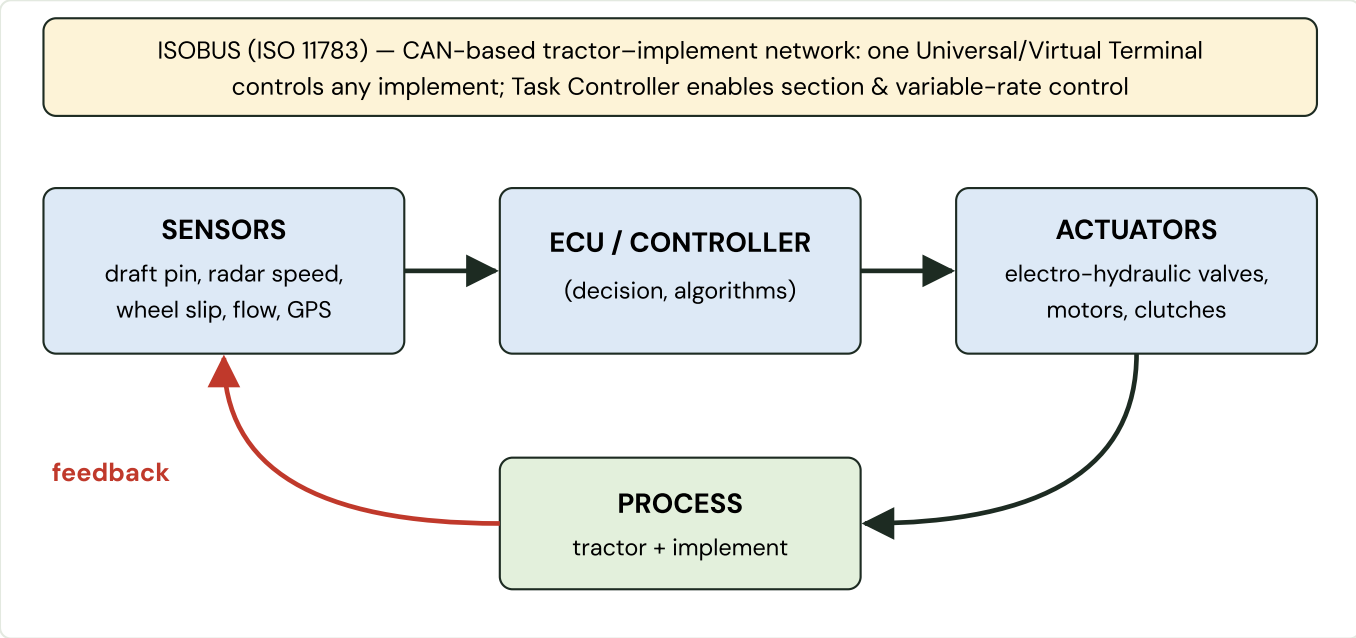
Mechatronics in Farm Machinery

Syllabus topic: Machine learning, IoT, robotics and mechatronics in farm machinery

Mechatronics = synergistic integration of mechanics + electronics + computing/control in one system.

The control loop

1. Sensors measure.
2. An ECU (electronic control unit) decides.
3. Actuators act.
4. Feedback closes the loop.



Mechatronic control loop on modern tractors (redrawn from the source figure)

Typical tractor sensors: draft load pins in lower links (electro-hydraulic hitch control), radar/GPS true ground speed, wheel speed (slip computation), engine & PTO speed, fuel flow, hydraulic pressure, seed flow/blockage sensors, grain-yield and moisture sensors on combines.

CAN bus and ISOBUS

- **CAN bus:** a two-wire serial network over which all ECUs share data (engine, transmission, hitch, implement).
- **ISOBUS (ISO 11783)** extends CAN to the tractor–implement interface: a standard connector + a Universal/Virtual Terminal in the cab controls ANY maker's ISOBUS implement. The Task Controller logs work and drives **section control** and **variable-rate** application (Unit 10 precision agriculture).

Electro-hydraulic controls

Solenoid/proportional valves replace mechanical linkages — e.g., EHC hitch control, auto-steer steering valves, header-height control on combines.

Practice MCQs

Q192. Mechatronics is the integration of:

(a) Only mechanics and hydraulics
(b) Mechanics, electronics and computing/control
(c) Chemistry and biology
(d) Civil and electrical works

Q193. The international standard for the tractor–implement electronic interface (ISOBUS) is:

(a) ISO 730
(b) ISO 11783
(c) ISO 9001
(d) ISO 500

Q194. ISOBUS communication is based on:

(a) Bluetooth
(b) CAN bus
(c) Wi-Fi
(d) RS-232 only

Q195. Draft on modern electro-hydraulic hitch tractors is sensed by:

(a) Spring balance
(b) Load-sensing pins in the lower links
(c) Operator feel
(d) Fuel gauge

Answers

Q192 — (b) Mechanics, electronics and computing/control

Q193 — (b) ISO 11783

Q194 — (b) CAN bus

Q195 — (b) Load-sensing pins in the lower links

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.